

6.20 NUMERICAL METHODS, NM (4776) AS

Objectives

To provide students with an understanding that many mathematical problems cannot be solved analytically but require numerical methods.

To develop a repertoire of simple numerical methods and to give experience in using them.

To state or prove theoretical results about the accuracy of these numerical techniques and to demonstrate the control of error in practice.

To implement these numerical methods on computers and to develop an awareness of the difficulties which can arise when computers are used to do mathematics.

Assessment

Examination (72 marks)
1 hour 30 minutes
The examination paper has two sections:

Section A: 5 - 7 questions, each worth at most 8 marks.
Section Total: 36 marks

Section B: two questions, each worth about 18 marks.
Section Total: 36 marks

Coursework (18 marks)
There is one assignment.

Assumed Knowledge

Candidates are expected to know the content of *C1* and *C2*.

Calculators

In the MEI Structured Mathematics specification, no calculator is allowed in the examination for *C1*. For all other units, including this one, a graphical calculator is allowed.

NUMERICAL METHODS, NM		
Specification	Ref.	Competence Statements

SOLUTION OF EQUATIONS

Bisection method. False Position (linear interpolation). Secant method. Fixed point iteration. Newton-Raphson method.	NMe1	Understand the graphical interpretations of these methods.
	2	Know how to solve equations to any required degree of accuracy using these methods.
	3	Understand the relative computational merits and possible failure of the methods.
	4	Know that fixed point iteration generally has first order convergence, Newton-Raphson generally has second order convergence.

ERRORS

Absolute and relative error.	NMv1	Know how to calculate errors in sums, differences, products and quotients.
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Error propagation by arithmetical operations and by functions.	2	Know how to calculate the error in $f(x)$ when there is an error in x .
	3	Understand the effects on errors of changing the order of a sequence of operations.
Errors in the representation of numbers: rounding; chopping.	4	Understand that computers represent numbers to limited precision.
	5	Understand the consequences of subtracting nearly equal numbers.

NUMERICAL DIFFERENTIATION

Forward difference method. Central difference method.	NMc1	Know how to estimate a derivative using the forward and central difference methods with a suitable value (or sequence of values) of h .
	2	Have an empirical and graphical appreciation of the greater accuracy of the central difference formula.

NUMERICAL INTEGRATION

Mid-point rule. Trapezium rule. Simpson's rule.	NMc3	Be able to evaluate a given definite integral to any desired degree of accuracy using these methods.
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The relationship between methods.	4	Know the order of errors of the midpoint and trapezium rules. Understand the development of Simpson's rule from the midpoint and the trapezium rules.
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NUMERICAL METHODS, NM			
Ref.	Notes	Notation	Exclusions

SOLUTION OF EQUATIONS

NMe1	Staircase and cobweb diagrams.		Proofs of orders of convergence.
2			
3			
4			

ERRORS

NMv1		Exact value: x Approximate value: X Absolute error: $X - x$ Relative error: $\frac{X - x}{x}$.	
2	$f(x+h) \approx f(x) + hf'(x)$		Functions of more than one variable.
3			
4			
5			

NUMERICAL DIFFERENTIATION

NMc1		$f'(x) \approx \frac{f(x+h) - f(x)}{h}$ $f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$	
2			

NUMERICAL INTEGRATION

Nmc3		M_n : midpoint rule with n strips T_n : trapezium rule with n strips S_{2n} : Simpson's rule with $2n$ strips.	
4	$T_{2n} = \frac{1}{2}(M_n + T_n)$		

NUMERICAL METHODS, NM		
Specification	Ref.	Competence Statements

APPROXIMATIONS TO FUNCTIONS		
Newton's forward difference interpolation method.	NMf1	Be able to use Newton's forward difference interpolation formula to reconstruct polynomials and to approximate functions.
Lagrange's form of the interpolating polynomial.	2	Be able to construct the interpolating polynomial of degree n given a set of $n + 1$ data points.

NUMERICAL METHODS, NM			
Ref.	Notes	Notation	Exclusions

APPROXIMATIONS TO FUNCTIONS

NMf1	Functions tabulated at equal intervals	$\Delta f(x) = f(x+h) - f(x)$
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Numerical Methods (NM) Coursework

Rationale

This module aims to develop skills in the areas of problem identification, use of numerical methods and control of error in practice. The coursework assignment should enable students to demonstrate a facility with technology and an awareness of the difficulties that can arise when computers are used to do mathematics.

Description

Students are expected to investigate a problem which is suitable for numerical solution, using one of the methods in the specification. (Problems which have analytical solutions are acceptable only if the analytical solution is too time-consuming, or too advanced, to be feasible.) Students should use a computer to develop a solution which is both efficient and accurate. In particular they must show how the desired accuracy has been achieved, either by means of sufficient iterations of the numerical process to ensure that the accuracy has been achieved, or by means of a theoretical analysis of errors.

Since it is assumed that the coursework will be implemented on a computer, the coursework will often arise naturally from the work done in the module.

The coursework counts for 20% of the total assessment for this unit.

Assessment

The task must be assessed on the coursework assessment sheets.

Numerical Methods (NM) Coursework: Assessment Sheet

Task: Candidates are expected to investigate a problem which is suitable for numerical solution, using one of the methods in the specification.

Coursework Title											
Candidate Name						Candidate Number					
Centre Number							Date				
Domain	Mark	Description						Comment		Mark	
Problem specification	1	An appropriate problem is identified.									
	1	There is a correct explanation of why it is appropriate for numerical solution.									
Strategy	1	An appropriate procedure is used.									
	1	The candidate explains why the procedure to be adopted is appropriate to the problem.									
Formula application	1	There is an application of routine algorithms and methods, which is at least partially correct.									
	1	There is a substantial application of routine algorithms and methods which is largely correct.									
	1	The application is fully correct.									
Use of technology	1	Appropriate use is made of calculator or computer software.									
	1	It is clearly explained.									
Error analysis	1	There is a competent analysis of errors, but it may be brief.									
	1	The analysis is developed for the particular problem, either from theoretical formulae or iteration of methods to achieve convergence.									
	1	This development leads to an improved solution.									
Interpretation	1	The solution is clearly expressed.									
	1	The solution is produced to a high degree of accuracy (e.g. 6 significant figures).									
	1	In addition there is a discussion of the validity of the solution.									
	1	Possible limitations are identified.									
Oral communication	2	Presentation		Please tick at least one box and give a brief report.							
		Interview									
		Discussion									
Half marks may be awarded but the overall total must be an integer. Please report overleaf on any help that the candidate has received beyond the guidelines								TOTAL		18	

Coursework must be available for moderation by OCR